

Aliah University
End-Term (Autumn) Examination 2020
Title of the Paper: Indian Constitution

Full Marks: 80

Time: 3 Hours

Section-I-Answer any two (2) of the following:

2X20=40

1. What is Preamble? Is it a Part of the Constitution? Can it be used for interpretation of the Constitution? Also explain its significance.
2. Critically examine the historical development of the Constitution of India.
3. Discuss the characteristic features of the Indian Constitution.

Section-II- Answer any Three (3) of the following:

3X10=30

1. "The Directive Principles of State Policy constitute a comprehensive political, social and economic program for a modern democratic welfare State". Examine.
2. Discuss in detail the 'Fundamental Duties'.
3. Discuss in brief the various writs that can be issued by the Supreme Court of India.
4. Write a short note on "Parliamentary System".

Section-III- Answer any Two (2) of the following:

2X5=10

1. Discuss in detail the nature of the "Indian Parliament".
 2. Write a short note on the 'Fundamental Rights'.
 3. In what sense the Constitutional Amendments has provided significance steps for the Citizens of India?
-

Aliah University

End-Term Examination, March 2021

Course: B. Tech. (Civil Engineering)

Year & Semester: 2nd Yr. (III Sem.)

Subject: Mechanics of Materials

Subject Code: CENUGPC01

Full Marks: 80

Time: 3 hrs.

Answer any four questions. All questions carry equal marks.

1. (a) Discuss the stress-strain curve and behavior of mild steel under tensile test.

(b) A rectangular block of a material is subjected to a tensile stress of 50 N/mm^2 on one plane and a tensile stress of 40 N/mm^2 on a plane right angle, together with shear stress of 30 N/mm^2 on the same planes. Find the direction of principle planes and magnitude of principle stresses.

[10+10=20]

2. (a) What is a flitched beam? How moment of resistance gets enhanced in this beam? Explain.

(b) A flitched beam consists of a wooden joist 100 mm wide and 160 mm deep strengthened by steel plates 6 mm thick, 160 mm deep attached both the sides. If the maximum stress in timber is 5 N/mm^2 . Find maximum stress in steel and moment of resistance of the section. (Take modular ratio, $m=20$)

[10+10=20]

3. A steel beam of *I* section is 500 mm deep. Each flange is 200 mm wide and 25 mm thick. The web is 15 mm thick. The beam section is subjected to 400 kN. Determine the shear stress distribution for the beam section.

[20]

4. (a) Explain Euler's theory and assumptions made in this theory. Also explain effective length of columns for different end conditions.

(b) A vertical steel pipe 4 m long having an external diameter of 100 mm and an internal diameter of 75 mm supports a 2 m wide (from centre of the pipe), 0.6 m deep vertical sign board at top and the pipe is fixed at bottom. Determine the maximum bending and torsional stress for the pipe due to wind pressure 2500 N/mm^2 .

[10+10=20]

5. (a) What are hoop stresses and longitudinal stresses? Explain.

For a city water supply, the diameter of pipe to be used is 1.0 m and head of water is 60 m. Find the thickness of cast iron pipe required, if the permissible tensile stress in cast iron is 20 N/mm^2 .

(b) Explain moment area methods for determining slope and deflection in beams.

Find slope at supports and deflection at mid span in a simply supported beam carrying a point load at mid span using this method.

[10+10=20]

Aliah University

Department of Civil Engineering

B.Tech. (2nd Year) Final Semester Examination 2021

Sub : Introduction to Civil Engineering (CENUGHU01)

Full Marks : 80

Time: 3 hr

Answer all questions. Figures in the margin indicate full marks.

1. What are the advantages and disadvantages of smart cities ? (8)
2. What are the tests conducted on plain cement concrete? (4)
3. How energy efficiency building is planned? (6)
4. How depth and width for a wall footing are determined? (6)
5. What is guniting, shortcreting, shoring and underpinning ? (4)
6. Write down the advantages of lean construction. (6)
7. Write down the components of basic waves. (5)
8. Write the name of temporary structure used in construction. (5)
9. What are the factors considered for planning of harbour ? (5)
10. Describe the function of different ingredients in plain cement concrete? (10)
11. What are the principle modes of sediment transport in natural stream. (5)
12. Describe the mechanism: how corrosion takes place in steel in reinforced concrete ? (6)

13. Name any two IS codes: (i) for rural road design (ii) design of hydraulic structure and (iii) water supply with sewage disposal. (5)
14. For staircase planning (of a residential building of floor to floor height 3 m) what are the desirable relationship between tread and riser. Draw the plan (not in scale) of the stair case for 4.5 m length and 3.0 m width. (5)

ALIAH UNIVERSITY
DEPARTMENT OF MATHEMATICS AND STATISTICS
PAPER MATUGBS03 : ENGINEERING MATHEMATICS 3

Unless otherwise stated, notations carry their usual meanings.

Time-3 Hours

Full Marks-80

Group-A

1. Answer any **TEN** questions:

$10 \times 2 = 20$

- (a) Write the relation between the operators Δ and E . Also, prove it.
- (b) Define Absolute error and Relative error in context of numerical problems.
- (c) Prove that

$$1 + \delta^2 \mu^2 = \left(1 + \frac{\delta^2}{2}\right)^2.$$

- (d) What do you mean by Interpolation?
- (e) Find the missing term from the following table using operators:

$X :$	0	1	2	3	4
$Y :$	1	3	9	?	81.
- (f) If A and B are independent events then prove that the events A and B^c are also independent.
- (g) Define conditional probability.
- (h) Prove that the probability of any event lies between 0 and 1.
- (i) Define Sample space and Event of a random experiment.
- (j) The mean and variance of a Binomial (n, p) distribution are respectively 20 and 16. Find the values of n and p .
- (k) Find the Mean and Variance of the first n natural numbers.
- (l) Find the Mean deviation of the numbers 31, 35, 29, 63, 56, 72, 37, 40, 32, 63 from the mean.
- (m) If a constant K is added with each of n observations, how the mean of the observation changes?
- (n) The variates x and y are connected by $y = 4x + 5$ and if Standard Deviation of x is 2. Find the Standard Deviation of y .
- (o) Define Raw moments and Central moments.

Group-B

Answer any **TWO** from **2-4**

2. (a) From the following table of values x and $f(x)$ determine the value of $f(0.29)$ using Newton's backward interpolation formula.

$x :$	0.20	0.22	0.24	0.26	0.28	0.30
$f(x) :$	1.6596	1.6698	1.6804	1.6912	1.7024	1.7139.

[5]

(b) Find $f(1.2)$ where $f(0) = 1, f(1) = 1.5, f(2) = 2.2, f(3) = 3.1, f(4) = 4.9$. [5]

3. (a) Find the Cubic polynomial which takes the following values:

$$y(1) = 24, y(3) = 120, y(5) = 336 \text{ and } y(7) = 720.$$

Hence or otherwise, obtain the value of $y(8)$. [5]

(b) From the following table of values of x and $f(x)$ obtain $f'(x)$ and $f''(x)$ for $x = 1.6$.

$x :$	1.0	1.1	1.2	1.3	1.4	1.5	1.6
$f(x) :$	1.2661	1.3262	1.3937	1.4693	1.5534	1.6647	1.7500.

[5]

4. (a) Evaluate $\int_0^3 (2x - x^2)dx$, taking 6 intervals by Trapezoidal rule. [5]

(b) Find the root of the equation $5x^3 - 20x + 3 = 0$ lying on the interval $[0, 1]$ with an accuracy of 10^{-4} . [5]

Group-C

Answer any **TWO** from **5-7**

5. (a) For any two events A and B prove that [5]

$$P(A \cup B) = P(A) + P(B) - P(A \cap B).$$

(b) An urn contains 6 red and 4 Black balls, two balls are drawn at random. Find the probability that

i. Both the balls are red.

ii. One ball is red and another ball is black.

iii. Both of them are of the same colour. [5]

6. (a) A speaks the truth in 75% cases and B in 80% cases. In what percentage of cases are they likely to contradict each other in stating the same fact? [3]

(b) I have 6 gold coins, 4 silver coins and 3 bronze coins. I have taken out 3 coins at random. What is the probability that they are all of identical material? What is the probability that they are all of the same material? [2+2]

(c) A card is drawn at random from a pack of well shuffled cards. What is the probability that it is a heart or a queen? [3]

7. (a) There are two identical boxes. The 1st box contains 5 white, 7 red balls and the second box contains 5 white, 5 red balls. One box is chosen at random and a ball is drawn from it. If the ball drawn is found to be white. Calculate the prob. that it is drawn from the 1st box. [5]

(b) If X and Y be two random variables then prove that

i. $E(X + Y) = E(X) + E(Y)$,

ii. $\text{Var}(X + Y) = \text{Var}(X) + \text{Var}(Y)$, if X and Y are independent. [2+3]

Group-D

Answer any **TWO** from **8-10**

8. (a) Calculate the Median and Mode of the following distribution:
Height (in inch): 56–60 61–65 66–70 71–75 76–80
No. of Persons: 8 24 44 27 7. [5]
- (b) Find the Mean and Standard deviation of the following distribution: [5]
Weight (in lb): 118–126 127–135 136–144 145–153 154–162 163–171 172–180
No. of Students: 3 5 9 12 5 4 2.
9. (a) Define Population, Sample and Frequency distribution. [5]
- (b) i. Prove that correlation coefficient is the geometric mean of two regression coefficient.
ii. Find the Mean, Median and Mode for the set of values 3, 5, 2, 6, 5, 9, 5, 2, 8, 6. [2+3]
10. (a) Prove that the Arithmetic Mean of the two regression coefficient is always greater than the coefficient of correlation. [4]
- (b) Prove that $\text{Var}(x) = Ex(x-1) - m(m-1)$; where, $m = E(x)$. [3]
- (c) For the given data, calculate the coefficient of variation. Pearson's measure of skewness = 0.42; Arithmetic mean = 86; Median = 80. [3]
-

ALIAH UNIVERSITY

End-Semester Examination- March, 2021

Year: 2nd Year (Dept. of Civil Engg.) Semester: 3rd Semester
Subject: Surveying and Geomatics Subject Code: CENUGPC02
Full Marks: 80 Time: 3 hr.

Attempt any **five** questions and assume suitable data if required.

- 1 (a) What do you mean by remote sensing? Discuss the basic process of data acquisition and data analysis. 8
(b) What do you mean by ideal and real characteristics of remote sensing system? Discuss in details. 8
- 2 (a) Discuss the Energy interaction in atmosphere and the Energy interaction with earth surface feature. 8
(b) Discuss the Airborne and Space-borne Remote Sensing. 8
- 3 (a) Discuss the spatial, Spectral, temporal and radiometric resolution. 8
(b) Discuss the geo-synchronous, sun-synchronous and polar- synchronous orbit. 8
- 4 (a) What do you mean by surveying and its objective? Discuss the primary and other division of surveying. 8
(b) The true length of a line was known to be 360 m. The line was measured with a 30 m tape and found to be 361.5 m. What was correct length of the tape? 8
- 5 (a) Discuss the following terms as: Bearing of line, True meridian, Magnetic meridian, arbitrary meridian, designation of bearings and local attraction. 8
(b) Compute the angle between the lines AB and AC, if their respective bearings are (i) 30° 15' and 78° 30' (ii) 28° 45' and 137° 15' (iii) 76° 45' and 248° 30' and (iv) 52° 30' and 328° 45' 8
- 6 (a) Discuss the various types of tape and chain used in surveying? 8
(b) Discuss the following terms: Horizontal plane and horizontal line, Vertical plane and vertical line, datum surface, elevation, bench mark, back sight, intermediate sight and fore sight. 8

ALIAH UNIVERSITY

B. Tech. Semester:III, Civil Engineering

Odd Semester Examination 17th March-2021

Subject Code &Name: (CENUGPC03)(CE205), Building Materials

Time:3 Hours

Total Marks: 80

- Instruction:** (1) Figures to the right indicate full marks.
(2) Assume suitable data if required.
(3) Attempt any five questions.
(4) Write the detail address of your local area.

- Que. – 1** Describe the process of burning of bricks in your local area where you are living. Write your detail address so that it can be verified. **16**
- Que. – 2** Write the name of river nearer to you. And discuss the characteristics of sand which is available in that river. Compare these characteristics with good quality of sand. **16**
- Que. – 3** Enlist and explain the eight historical structures which are constructed with lime mortar. **16**
- Que. – 4** How is the cement classified? Describe the wet process of manufacturing cement. **16**
- Que. – 5** Which grade of concrete is used while construction of your home? Also write details notes on classification of concrete. **16**
- Que. – 6** Prepare at least one poster on given topic. Which explain every aspects of the given topic? You can prepare poster either pen pencil or digitally. Attach this poster with your answer sheet in pdf format. **16**
- Bricks, Aggregates, Lime, Cement, Concrete, Mortar, Woods, Paints.

END OF PAPER